OPIM-5671 Team Project #2: Collecting and Reading Massive Texts with Machines

Overall Project Guidance

October 27, 2026

Documented by **Jaeung Sim**

1. Objective

Apply methods taught in class to a Kaggle text corpus and develop credible empirical findings with practical business or public-interest value.

2. Project Overview

Item	Requirement
Project focus	Open-ended, question-driven text analysis with clear business or public value.
Data	A publicly accessible Kaggle dataset containing original textual observations.
Technical level	One primary course-related text-mining technique plus one simple supporting analysis. Advanced alternatives are optional, not automatically rewarded.
Computing expectation	Reasonable-size data and free resources such as Google Colab. Paid tools are not required.
Presentation	Tue, Nov 17: strictly 10 minutes + 2 minutes of Q&A; no prototype or separate report.

3. Necessary Components

- **Research question:** State a focused question, intended decision-maker, and why it matters.
- **Data reasoning:** Describe the text corpus, source, scope, preprocessing, and possible biases.
- **Methodological judgment:** Select a suitable primary text-mining technique and complementary analysis.
- **Empirical evidence:** Produce at least two substantive findings supported by figures, comparisons, or model outputs.
- **Interpretation and validation:** Check what the outputs mean and distinguish evidence from speculation.
- **Decision value:** Translate findings into realistic business or public-interest implications.

4. Deliverables and Schedule

Deliverable	Due	How to submit	Minimum contents
Team proposal	Mon, Nov 2, 11:59 PM	Team email; cc all	Question, user/decision, Kaggle link, method plan, feasibility.
Final notebook	Sun, Nov 15, 11:59 PM	Team email; cc all	Reproducible processing, primary/support methods, findings, and checks.
Final slide deck	Sun, Nov 15, 11:59 PM	Team email; cc all	Question, data, methods, 2+ findings, implications, limitations, contributions.

Submitted slide decks will be shared with all classmates for educational purposes. Exclude sensitive or unauthorized content. For large datasets, provide Kaggle links instead of files.

5. Technical Expectations

- Use at least one primary technique from class (e.g., sentiment analysis, emotion extraction, LDA topic modeling, TF-IDF/SVD, or text-based prediction).
- Include one simple supporting analysis, such as word frequencies, comparisons across groups/time, or inspection of representative texts.
- Explain key preprocessing choices and why the method suits the research question.
- Present at least two substantive empirical findings, each linked to supporting quantitative evidence.
- Validate important scores, topic labels, or interpretations through representative texts or other sensible checks.
- Distinguish observed associations from causal claims; acknowledge uncertainty and relevant limitations.
- More advanced methods are welcome when feasible, but complexity, number of models, and expensive tools earn no automatic advantage.

6. Data and Computing Constraints

- Use a publicly accessible Kaggle-hosted dataset with actual raw text; cite its Kaggle page and original source/license when available.
- Describe the document unit, corpus size, relevant coverage, missingness, and likely selection bias.
- Use a justified subset if needed; a larger dataset is not inherently a better analysis.
- The analysis should run with free Google Colab or comparable resources; paid APIs/GPUs and scraping are not expected.
- Teams may select the same dataset but must develop their own question, comparisons, and interpretations.

7. Empirical Findings and Interpretation

The main deliverable is a defensible analytical story, not a prototype. Present at least two substantive findings; for each, show:

- **Evidence:** A clear figure, quantitative comparison, or other reproducible result.
- **Interpretation:** What the result suggests about the research question, supported by representative texts when useful.
- **Implication:** How the finding might inform a specific decision or public-interest issue.
- **Credibility:** Relevant checks, uncertainty, competing explanations, or dataset/method limitations.
- **Originality:** Own research questions and thoughtful comparisons are encouraged, without a separate originality score.

8. Presentation Requirements

- **10 minutes (strict):** Present the question/value, data/methods, key evidence and interpretation, recommendations, and limitations. Most time should focus on findings.
- **2 minutes:** Q&A. Every member should be prepared to explain the dataset, analysis, and conclusions.
- **Slides:** Use readable figures and minimal code; include a slide identifying each team member's contribution.
- **AI use:** AI tools may assist with code, analysis, or presentation preparation. Teams must verify claims, explain their work, and avoid reading scripts verbatim.

9. Audience Review Reports

Each student submits a brief review after every other team's presentation using the class form:

- What is one meaningful finding you learned?
- Which interpretation or practical implication was most interesting or useful?
- What question or constructive suggestion would you offer the team?

10. Teammate Evaluation

Each member completes a confidential qualitative review of teammates. Consistent, credible evidence of substantial contribution gaps, free-riding, unethical conduct, or serious team problems may affect individual evaluation; minor stylistic preferences will not.

11. Evaluation Dimensions

Dimension	What strong work demonstrates	Primary evidence	Weight
Methodological Soundness	A suitable primary method and simple supporting analysis are selected and justified.	Notebook + slides	15%
Analytical Execution	Correct, readable, reproducible Python workflow, including data preparation and useful validity checks.	Notebook	15%
Business/Public Value	A meaningful problem and practical implications for a clearly identified decision-maker.	Slides	15%
Interpretation & Insights	At least two well-supported findings, careful interpretation, credibility checks, and acknowledged limitations.	Slides + notebook	30%
Communication & Presentation	A clear, concise narrative, effective evidence displays, and professional Q&A.	Slides + Q&A	15%
Team Contribution	Responsible collaboration; substantiated contribution problems may affect individual evaluation.	Confidential peer review	10%

12. Team Proposal Template

- **Project title:** A concise working title.
- **Kaggle dataset:** Dataset link, original source (if known), number/type of text observations, and relevant variables.
- **Research question:** What will you investigate, and why is the question important?
- **Intended user / decision:** Who can use the findings, and for what decision?
- **Analytical plan:** One primary text-mining technique and one simple supporting analysis, with brief justification.
- **Scope check:** Evidence that the text can be loaded and analyzed with free computing resources; potential data limitations.

Keep the proposal brief (about one page). It is a feasibility checkpoint, not a commitment to specific findings; reasonable method adjustments are allowed after feedback.